Data Technician

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Day 2: Task 1

It is a common software development interview question to create the below with a certain programming language. Create the below using Python syntax, test it and past the completed syntax and output below.

FizzBuzz:

Go through the integers from 1 to 100.
If a number is divisible by 3, print "fizz."
If a number is divisible by 5, print "buzz."
If a number is both divisible by 3 and by 5, print "fizzbuzz."
Otherwise, print just the number.

Paste your completed work to the right

```python
for n in range(1,101):
    if n % 3 == 0 and n%5==0:
        print("FizzBuzz")
    elif n % 3 == 0:
        print("fizz")
    elif n % 5 == 0:
        print("Buzz")
    else:
        print(n)
```

```
43
44
FizzBuzz
46
47
fizz
49
Buzz
fizz
52
53
fizz
Buzz
56
fizz
58
59
FizzBuzz
61
62
fizz
64
Buzz
```

```
fizz
67
68
fizz
Buzz
71
fizz
73
74
FizzBuzz
76
77
fizz
79
Buzz
fizz
82
83
fizz
Buzz
86
fizz
88
89
FizzBuzz
91
92
fizz
94
Buzz
fizz
97
98
fizz
```

Day 3: Task 1

Using the 'student.csv' which can be downloaded here, complete the below exercises as a group and paste your input and output. Although this is a group activity, everyone should have the below answered so it supports your portfolio:

Exercise 1: Loading and Exploring the Data

1. Question: "Write the code to read a CSV file into a Pandas DataFrame."
2. Question: "Write the code to display the first 5 rows of the DataFrame."
3. Question: "Write the code to get the information about the DataFrame."
4. Question: "Write the code to get summary statistics for the DataFrame."

```
1. Import pandas as pd
   Df = pd.read_csv('students_csv')

   2. Import.pandas as pd
   Df = pd.read_csv("students.cv")
   #Display first 5 rows
   Result=Df.head(5)
   Print("First 5 rows of the Data Frame:")
   Print(Result)

3. Df.info()
```

Exercise 2: Indexing and Slicing

1. Question: "Write the code to select the 'name' column."
2. Question: "Write the code to select the 'name' and 'mark' columns."
3. Question: "Write the code to select the first 3 rows."
4. Question: "Write the code to select all rows where the 'class' is 'Four'."

```
1. Df[name]
2. Selected_columns = df["name", "mark"]
   Print(Selected_columns.head()
```

3. Display df.head(3)
4. Select class_four_df.head()

Exercise 3: Data Manipulation

1. Question: "Write the code to add a new column 'passed' that indicates whether the student passed (mark >= 60)."
2. Question: "Write the code to rename the 'mark' column to 'score'."
3. Question: "Write the code to drop the 'passed' column."

1. Df('passed') = df('mark') >=60
   Display(df.head())
2. Df.rename_columns=('mark':'score')
   Display(df.head())
3. Df.drop('passed')

Exercise 4: Aggregation and Grouping

1. Question: "Write the code to group the DataFrame by the 'class' column and calculate the mean 'mark' for each group."
2. Question: "Write the code to count the number of students in each class."
3. Question: "Write the code to calculate the average mark for each gender."

1. mean_score_by_class=df.groupby("class") ('score') . Mean()
Print(mean_score by class)
2. Students_per_Class=df.groupby("class_size")
   Print(students_per_class)

3. Avg_score_by gender= df.groupby(@gender') , ("score").mean()

```
Print(avg_score_by_gender)
```

Exercise 5: Advanced Operations

1. Question: "Write the code to create a pivot table with 'class' as rows, 'gender' as columns, and 'mark' as values."
2. Question: "Write the code to create a new column 'grade' where marks >= 85 are 'A', 70-84 are 'B', 60-69 are 'C', and below 60 are 'D'."
3. Question: "Write the code to sort the DataFrame by 'mark' in descending order."

```
1. Pivot _table_ class _gender _score= pd._pivot _table(df,values='score' ,
   index='class' clums='gender'
   Print(Pivot_table_class_gender_score

2. I'm not sure

3. I'm not sure
```

Exercise 6: Exporting Data

1. Question: "Write the code to save the DataFrame with the new 'grade' column to a new CSV file."

```
1. df.to_csv('students_with_grades.csv',) where index=False
Print(data frame saved to students_with_grades_csv)
```

Exercise 7: If finished early try visualising the results

Day 4: Task 1

Using the 'GDP (nominal) per Capita.csv' which can be downloaded here, complete the below exercises and paste your input and output. Work individually, but we will work and support each other in the room.

- Read and save the 'GDP (nominal) per Capita' data to a data frame called "df" in Jupyter notebook

- Print the first 10 rows
- Print the last 5 rows
- Print 'Country/Territory' and 'UN_Region' columns

1.

```
import pandas as pd
df = pd.read_csv('GDP (nominal) per Capita.csv')
display(df.head())
```

	Unnamed: 0	Country/Territory	UN_Region	IMF_Estimate	IMF_Year	WorldBank_Estimate	WorldBank_Year	UN_Estimate	UN_Ye
0	1	Monaco	Europe	0	0	234316	2021	234317	20
1	2	Liechtenstein	Europe	0	0	157755	2020	169260	20
2	3	Luxembourg	Europe	132372	2023	133590	2021	133745	20
3	4	Ireland	Europe	114581	2023	100172	2021	101109	20
4	5	Bermuda	Americas	0	0	114090	2021	112653	20

2.

```
display(df.head(10))
```

	Unnamed: 0	Country/Territory	UN_Region	IMF_Estimate	IMF_Year	WorldBank_Estimate	WorldBank_Year	UN_Estimate	UN_Year
0	1	Monaco	Europe	0	0	234316	2021	234317	2021
1	2	Liechtenstein	Europe	0	0	157755	2020	169260	2021
2	3	Luxembourg	Europe	132372	2023	133590	2021	133745	2021
3	4	Ireland	Europe	114581	2023	100172	2021	101109	2021
4	5	Bermuda	Americas	0	0	114090	2021	112653	2021
5	6	Norway	Europe	101103	2023	89154	2021	89242	2021
6	7	Switzerland	Europe	98767	2023	91992	2021	93525	2021
7	8	Singapore	Asia	91100	2023	72794	2021	66822	2021
8	9	Isle of Man	Europe	0	0	87158	2019	0	0
9	10	Cayman Islands	Americas	0	0	86569	2021	85250	2021

3.

```
display(df.tail(5))
```

	Unnamed: 0	Country/Territory	UN_Region	IMF_Estimate	IMF_Year	WorldBank_Estimate	WorldBank_Year	UN_Estimate	UN_Year
218	219	Malawi	Africa	496	2023	635	2021	613	2021
219	220	South Sudan	Africa	467	2023	1072	2015	400	2021
220	221	Sierra Leone	Africa	415	2023	480	2021	505	2021
221	222	Afghanistan	Asia	611	2020	369	2021	373	2021
222	223	Burundi	Africa	249	2023	222	2021	311	2021

4.

```
display(df[['Country/Territory', 'UN_Region']])
```

	Country/Territory	UN_Region
0	Monaco	Europe
1	Liechtenstein	Europe
2	Luxembourg	Europe
3	Ireland	Europe
4	Bermuda	Americas
...	...	...
218	Malawi	Africa
219	South Sudan	Africa
220	Sierra Leone	Africa
221	Afghanistan	Asia
222	Burundi	Africa

Day 4: Task 2

Back with 'GDP (nominal) per Capita'. As a group, import and work your way through the Day_4_Python_Activity.ipynb notebook which can be found here. There are questions to answer, but also opportunities to have fun with the data – paste your input and output below.

Once complete, and again as a group, work with some
more data and have some fun –there is no set agenda for this section, other than to embed the skills developed this week. Paste your input and output below and upon return we'll discuss progress made.

Additional data found here. My Actual work for day 4 task 2 is in the email I sent to you. What you will see below are just some of the input and output when going through the questions that I found interesting as to show you that I did make progress

1. Number of countries per region

```
region_country_counts = df.groupby('UN_Region')['Country/Territory'].nun
region_country_counts.columns = ['UN_Region', 'Number of Countries']
display(region_country_counts)
```

	UN_Region	Number of Countries
0	Africa	55
1	Americas	48
2	Asia	51
3	Europe	48
4	Oceania	20
5	World	1

```
europe_countries = df[df['UN_Region'] == 'Europe']['Country/Territory'].
print("Countries in Europe:")
for country in europe_countries:
    print(country)

Countries in Europe:
Monaco
Liechtenstein
Luxembourg
Ireland
Norway
Switzerland
Isle of Man
Iceland
Channel Islands
Faroe Islands
Denmark
Netherlands
Austria
Sweden
Finland
Belgium
San Marino
Germany
United Kingdom
France
Andorra
European Union[n 1]
Malta
Italy
Slovenia
```

2

3.

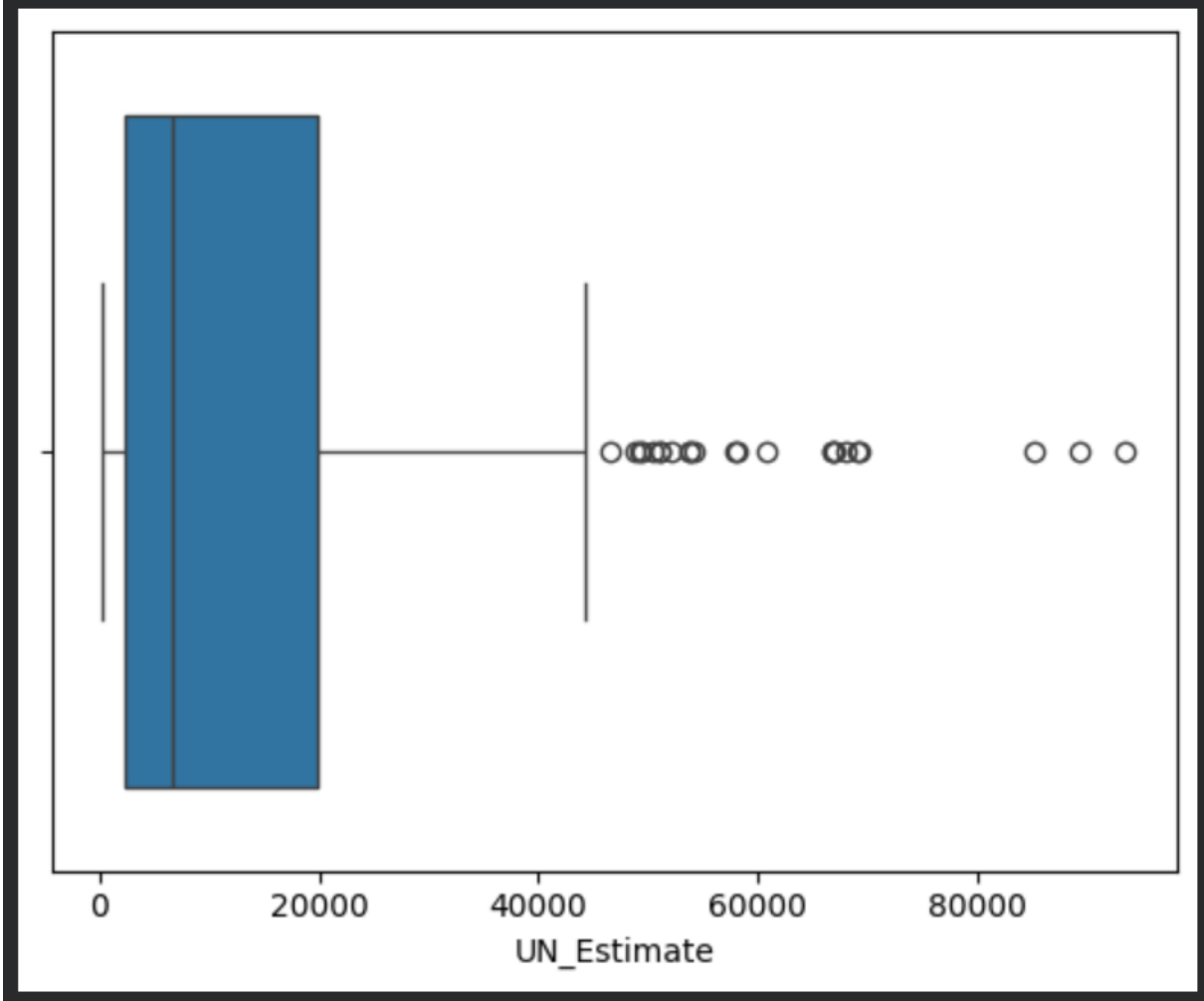
```
print("European Countries with IMF Estimate GDP Below European Average:")
for country in below_average_gdp_countries_df['Country/Territory']:
    print(country)
```

```
European Countries with IMF Estimate GDP Below European Average:
Malta
Italy
Slovenia
Czech Republic
Spain
Estonia
Lithuania
Portugal
Latvia
Slovakia
Greece
Croatia
Poland
Hungary
Romania
Bulgaria
Russia
Montenegro
Serbia
Bosnia and Herzegovina
Belarus
North Macedonia
Albania
Moldova
Kosovo
Ukraine
```

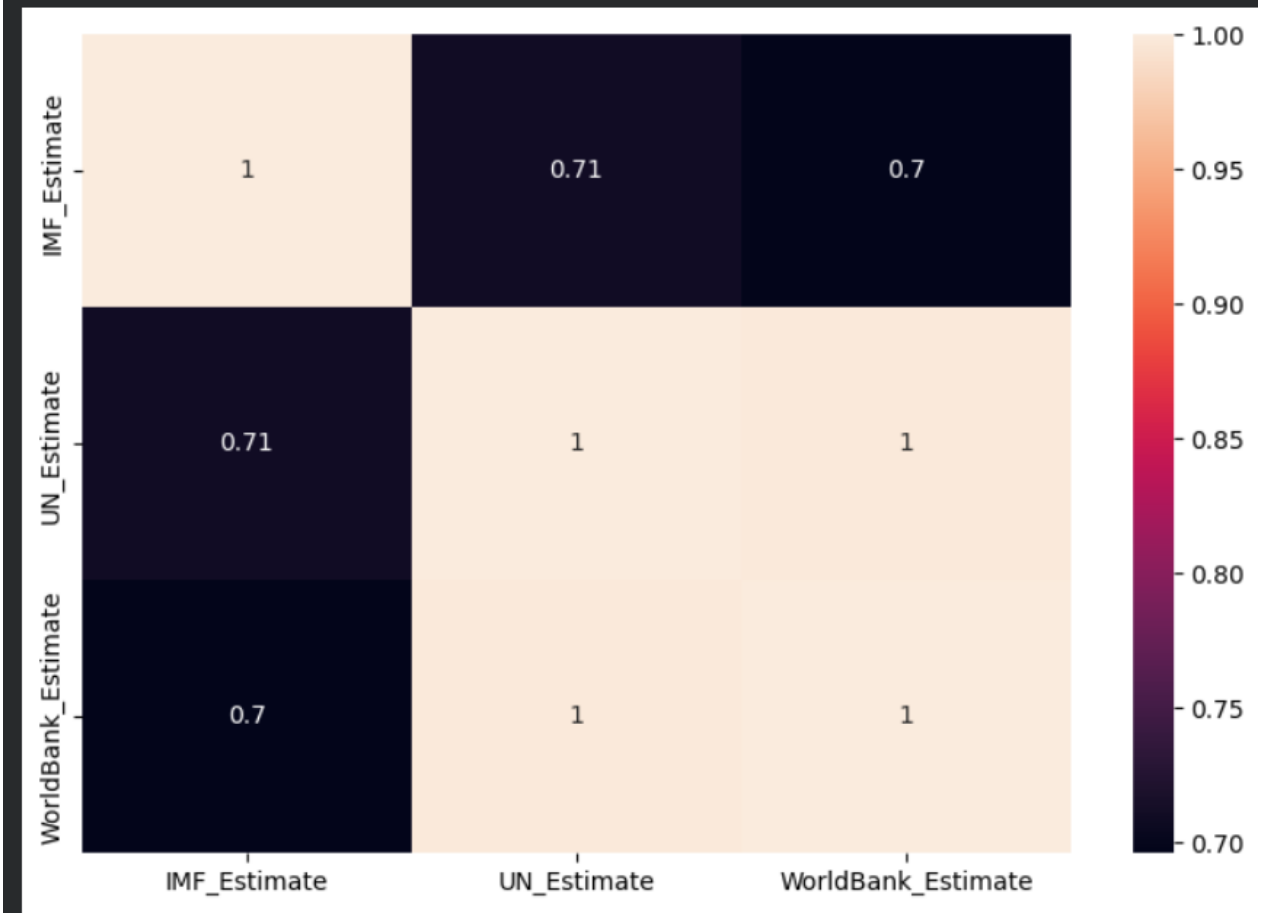
```
import numpy as np
df.replace(0, np.nan, inplace=True)
display(df.head())
```

	Unnamed: 0	Country/Territory	UN_Region	IMF_Estimate	IMF_Year	WorldBa
0	1	Monaco	Europe	NaN	NaN	
1	2	Liechtenstein	Europe	NaN	NaN	
2	3	Luxembourg	Europe	132372.0	2023.0	
3	4	Ireland	Europe	114581.0	2023.0	
4	5	Bermuda	Americas	NaN	NaN	

```
sns.boxplot(x=data["UN_Estimate"])
plt.show()
```



```
plt.figure(figsize=(9,6))
sns.heatmap(correlation_matrix, annot=True)
plt.show()
```



Course Notes

It is recommended to take notes from the course, use the space below to do so, or use the revision guide shared with the class:

We have included a range of additional links to further resources and information that you may find useful, these can be found within your revision guide.

END OF WORKBOOK

Please check through your work thoroughly before submitting and update the table of contents if required.

Please send your completed work booklet to your trainer.